



**Database System Final Project**  
**(Car Showroom Management System)**  
**(BS-SE-F23 Morning)**  
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# *(Car Showroom Management System)*

## *(System Introduction)*

### *1. Working*

*The Car Showroom Management System automates and streamlines the operations of a car showroom. It manages customer information, car inventory, bookings, test drives, sales transactions, warranties, and insurance. By simplifying processes, the system improves business efficiency, reduces manual workload, and enhances customer satisfaction. It also supports decision-making through real-time insights and comprehensive reporting capabilities.*

### *2. Key Features of the System*

- *Maintains detailed records of customers and their interactions with the showroom.*
- *Manages car inventory with attributes like model, price, availability, and specifications.*
- *Processes sales transactions, including customer details, payment methods, and delivery schedules.*
- *Generates reports based on sales, inventory, and other key metrics to support strategic decision-making.*

### *3. Description*

*This system stores data in relational tables with keys and constraints, ensuring consistency. Each table represents a key business object such as a customer, a car, or a sale. Relationships are managed using foreign keys, ensuring accurate record linkage across the database.*

### *4. Purpose*

*The purpose of this system is to simplify operations, provide accurate record-keeping, and enhance customer satisfaction. It supports business growth by automating tasks and offering insights into performance, inventory, and customer engagement.*

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## *(Entities and Their Relations)*

### *1. Entities and Attributes*

#### *1. Car*

- *Primary key: CarID*

- *Attributes: Make, Model, Year, Colour, EngineType, Transmission, Mileage, Price, VIN, Status*

## 2. Customer

- *Primary key: CustomerID*
- *Attributes: Name, Phone, Email, Address*

## 3. Salesperson

- *Primary key: SalespersonID*
- *Attributes: Name, Phone, Email, HireDate, Position, Salary*

## 4. Sale

- *Primary key: SaleID*
- *Attributes: DateOfSale, FinalPrice, PaymentMethod, CarID (Foreign Key), CustomerID (Foreign Key), SalespersonID (Foreign Key)*

## 5. TestDrive

- *Primary key: TestDriveID*
- *Attributes: ScheduledDate, Duration, Feedback, CarID (Foreign Key), CustomerID (Foreign Key), SalespersonID (Foreign Key)*

## 6. ShowRoom

- *Primary key: ShowroomID*
- *Attributes: Name, Location, Phone, ManagerName*

## 7. Inventory

- *Primary key: InventoryID*
- *Attributes: ItemName, Quantity, Category, ShowroomID (Foreign Key)*

## 8. Staff

- *Primary key: StaffID*
- *Attributes: Name, Role, ShowroomID (Foreign Key), Contact*

## 9. Payroll

- *Primary key: PayID*
- *Attributes: Salary, PayDate, StaffID (Foreign Key)*

## 10. Administration

- *Primary key: AdminID*
- *Attributes: Name, Contact, Role, ShowroomID (Foreign Key)*

## 11. Insurance

- *Primary key: InsuranceID*
- *Attributes: CarID (Foreign Key), CustomerID (Foreign Key), InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount*

## 12. Booking

- *Primary key: BookingID*
- *Attributes: CustomerID (Foreign Key), CarID (Foreign Key), BookingDate, BookingStatus, AdvanceAmount*

## 13. Warranty

- *Primary key: WarrantyID*
- *Attributes: CarID (Foreign Key), WarrantyStartDate, WarrantyEndDate, CoverageDetails*

## **2. Relationships**

1. **Sale and Car**  
*Relationship: Includes*
2. **Sales and Customer**  
*Relationship: Made by*
3. **Sales and Salesperson**  
*Relationship: Handled by*
4. **Booking and Customer**  
*Relationship: Made by*
5. **Booking and Car**  
*Relationship: Reserves*
6. **TestDrive and Customer**  
*Relationship: Requested by*
7. **TestDrive and Car**  
*Relationship: Taken for*
8. **TestDrive and Salesperson**  
*Relationship: Supervised by*
9. **Warranty and Car**  
*Relationship: Applies to*
10. **Insurance and Car**  
*Relationship: Covers*
11. **Insurance and Customer**  
*Relationship: Owned by*
12. **Payroll and Staff**  
*Relationship: Receives*
13. **Staff and Showroom**  
*Relationship: Works in*
14. **Administration and Staff**  
*Relationship: Manages*
15. **Administration and Showroom**  
*Relationship: Manages*



# *(Entity Relationship Diagram)*

## 1. ERD-Diagram

# *(System Design and Functionalities)*

## 1. Database Design

- Developed using both top-down and bottom-up design approaches.
- Includes an Entity Relationship Diagram (ERD) to visually represent relationships between entities such as Customers, Cars, Sales, and Staff.
  - Tables are defined with attributes, constraints, and relational keys to ensure consistency and data integrity.

## 2. Advanced Functionalities

- Custom Views: Enable simplified access to frequently needed information, such as available cars or customer bookings.
- Query-Based Reports: Generate real-time insights including total sales per month, top-performing salespersons, and car availability by showroom.

# *(Relational Schema)*

## 1. Top-Down-Approach

### 1. Car

- (CarID (PK), Make, Model, Year, Color, EngineType, Transmission, Mileage, Price, VIN, Status)

### 2. Customer

- (CustomerID (PK), Name, Phone, Email, Address)

### 3. Salesperson

- (SalespersonID (PK), Name, Phone, Email, HireDate, Position, Salary)

### 4. Sale

- (SaleID (PK), DateOfSale, FinalPrice, PaymentMethod, CustomerID (FK referencing Customer), CarID (FK referencing Car), SalespersonID (FK referencing Salesperson))

## **5. TestDrive**

- (*TestDriveID (PK), ScheduledDate, Duration, Feedback, CarID (FK referencing Car), CustomerID (FK referencing Customer), SalespersonID (FK referencing Salesperson)*)

## **6. ShowRoom**

- (*ShowroomID (PK), Name, Location, Phone, ManagerName*)

## **7. Inventory**

- (*InventoryID (PK), ItemName, Quantity, Category, ShowroomID (FK referencing Showroom)*)

## **8. Staff**

- (*StaffID (PK), Name, Role, Contact, ShowroomID (FK referencing Showroom)*)

## **9. Payroll**

- (*PayID (PK), Salary, PayDate, StaffID (FK referencing Staff)*)

## **10. Administration**

- (*AdminID (PK), Name, Contact, Role, ShowroomID (FK referencing Showroom)*)

## **11. Insurance**

- (*InsuranceID (PK), InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount, CarID (FK referencing Car), CustomerID (FK referencing Customer)*)

## **12. Booking**

- (*BookingID (PK), BookingDate, BookingStatus, AdvanceAmount, CustomerID (FK referencing Customer), CarID (FK referencing Car)*)

## **13. Warranty**

- (*WarrantyID (PK), WarrantyStartDate, WarrantyEndDate, CoverageDetails, CarID (FK referencing Car)*)

# **2. Bottom-Up-Approch**

## **5. 0NF**

*R ( CustomerID, Name, Phone, Email, Address, CarID, Make, Model, Year, Color, EngineType, Transmission, Mileage, Price, VIN, Status, SaleID, DateOfSale, FinalPrice, PaymentMethod, SalespersonID, SalespersonName, SalespersonPhone, SalespersonEmail, HireDate, Position, Salary, TestDriveID, ScheduledDate, Duration, Feedback, ShowroomID, ShowroomName, Location, ShowroomPhone, ManagerName, InventoryID, ItemName, Quantity, Category, StaffID, StaffName, StaffRole, StaffContact,*

*PayID, SalaryAmount, PayDate, AdminID, AdminName, AdminContact, AdminRole, InsuranceID, InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount, BookingID, BookingDate, BookingStatus, AdvanceAmount, WarrantyID, WarrantyStartDate, WarrantyEndDate, CoverageDetails )*

## **6. 1NF**

1. **Customer**  
(CustomerID, Name, Phone, Email, Address)
2. **Car**  
(CarID, Make, Model, Year, Color, EngineType, Transmission, Mileage, Price, VIN, Status)
3. **Salesperson**  
(SalespersonID, Name, Phone, Email, HireDate, Position, Salary)
4. **Sale**  
(SaleID, DateOfSale, FinalPrice, PaymentMethod, CustomerID, CarID, SalespersonID)
5. **TestDrive**  
(TestDriveID, ScheduledDate, Duration, Feedback, CarID, CustomerID, SalespersonID)
6. **Showroom**  
(ShowroomID, Name, Location, Phone, ManagerName)
7. **Inventory**  
(InventoryID, ItemName, Quantity, Category, ShowroomID)
8. **Staff**  
(StaffID, Name, Role, Contact, ShowroomID)
9. **Payroll**  
(PayID, SalaryAmount, PayDate, StaffID)
10. **Administration**  
(AdminID, Name, Contact, Role, ShowroomID)
11. **Insurance**  
(InsuranceID, InsuranceProvider, InsuranceStartDate, InsuranceEndDate, CoverageAmount, CarID, CustomerID)
12. **Booking**  
(BookingID, BookingDate, BookingStatus, AdvanceAmount, CustomerID, CarID)
13. **Warranty**  
(WarrantyID, WarrantyStartDate, WarrantyEndDate, CoverageDetails, CarID)

## **7. 2NF**

**Already in 2NF** — every non-key attribute depends on the full primary key of its table.

## **8. 3NF**

**Already in 3NF** — no attributes depend on other non-key attributes in the same table.

## **9. BCNF**

**Already in BCNF** — in all relations, **every determinant is a candidate key**, so no violations exist.

### 10.4NF

**Already in 4NF** — no relation contains attributes with **multivalued dependencies** not already tied to a candidate key.

### 11.5NF

**Already in 5NF** — no relation needs to be decomposed into **smaller projections** to eliminate anomalies or preserve information.

## (Table Description)

### 1. Customer

| Attribute  | Data Type | Size | Constraints |
|------------|-----------|------|-------------|
| CustomerID | Number    | -    | Primary Key |
| Name       | Varchar2  | 30   | Not Null    |
| Phone      | Varchar2  | 15   | Not Null    |
| Email      | Varchar2  | 30   | -           |
| Address    | Varchar2  | 50   | Not Null    |

### 2. Car

| Attribute    | Data Type | Size | Constraints                            |
|--------------|-----------|------|----------------------------------------|
| CarID        | Number    | -    | Primary Key                            |
| Make         | Varchar2  | 20   | Not Null                               |
| Model        | Varchar2  | 20   | Not Null                               |
| Year         | Number    | -    | -                                      |
| Color        | Varchar2  | 10   | -                                      |
| EngineType   | Varchar2  | 15   | -                                      |
| Transmission | Varchar2  | 10   | -                                      |
| Mileage      | Number    | -    | -                                      |
| Price        | Number    | -    | -                                      |
| VIN          | Varchar2  | 17   | Unique                                 |
| Status       | Varchar2  | 10   | Check(Status IN ('Available', 'Sold')) |

### 3. SalesPerson

| Attribute     | Data Type | Size | Constraints |
|---------------|-----------|------|-------------|
| SalespersonID | Number    | -    | Primary Key |
| Name          | Varchar2  | 30   | Not Null    |
| Phone         | Varchar2  | 15   | Not Null    |
| Email         | Varchar2  | 30   | -           |



|                 |          |    |                 |
|-----------------|----------|----|-----------------|
| <b>HireDate</b> | Date     | -  | Default SYSDATE |
| <b>Position</b> | Varchar2 | 20 | -               |
| <b>Salary</b>   | Number   | -  | -               |

#### 4. Sale

| Attribute            | Data Type | Size | Constraints                 |
|----------------------|-----------|------|-----------------------------|
| <b>SaleID</b>        | Number    | -    | Primary Key                 |
| <b>DateOfSale</b>    | Date      | -    | Default SYSDATE             |
| <b>FinalPrice</b>    | Number    | -    | -                           |
| <b>PaymentMethod</b> | Varchar2  | 15   | -                           |
| <b>CustomerID</b>    | Number    | -    | Foreign Key (CustomerID)    |
| <b>CarID</b>         | Number    | -    | Foreign Key (CarID)         |
| <b>SalespersonID</b> | Number    | -    | Foreign Key (SalespersonID) |

#### 5. TestDrive

| Attribute            | Data Type | Size | Constraints                 |
|----------------------|-----------|------|-----------------------------|
| <b>TestDriveID</b>   | Number    | -    | Primary Key                 |
| <b>ScheduledDate</b> | Date      | -    | -                           |
| <b>Duration</b>      | Varchar2  | 5    | -                           |
| <b>Feedback</b>      | Varchar2  | 100  | -                           |
| <b>CarID</b>         | Number    | -    | Foreign Key (CarID)         |
| <b>CustomerID</b>    | Number    | -    | Foreign Key (CustomerID)    |
| <b>SalespersonID</b> | Number    | -    | Foreign Key (SalespersonID) |

#### 6. ShowRoom

| Attribute          | Data Type | Size | Constraints |
|--------------------|-----------|------|-------------|
| <b>ShowroomID</b>  | Number    | -    | Primary Key |
| <b>Name</b>        | Varchar2  | 30   | Not Null    |
| <b>Location</b>    | Varchar2  | 50   | Not Null    |
| <b>Phone</b>       | Varchar2  | 15   | Not Null    |
| <b>ManagerName</b> | Varchar2  | 30   | -           |

#### 7. Inventory

| Attribute          | Data Type | Size | Constraints |
|--------------------|-----------|------|-------------|
| <b>InventoryID</b> | Number    | -    | Primary Key |
| <b>ItemName</b>    | Varchar2  | 25   | Not Null    |

|                   |          |    |                             |
|-------------------|----------|----|-----------------------------|
| <b>Quantity</b>   | Number   | -  | Check(Quantity >= 0)        |
| <b>Category</b>   | Varchar2 | 20 | Not Null                    |
| <b>ShowroomID</b> | Number   | -  | Foreign Key<br>(ShowroomID) |

## 8. Staff

| Attribute         | Data Type | Size | Constraints                 |
|-------------------|-----------|------|-----------------------------|
| <b>StaffID</b>    | Number    | -    | Primary Key                 |
| <b>Name</b>       | Varchar2  | 30   | Not Null                    |
| <b>Role</b>       | Varchar2  | 20   | Not Null                    |
| <b>Contact</b>    | Varchar2  | 15   | Not Null                    |
| <b>ShowroomID</b> | Number    | -    | Foreign Key<br>(ShowroomID) |

## 9. PayRoll

| Attribute           | Data Type | Size | Constraints           |
|---------------------|-----------|------|-----------------------|
| <b>PayID</b>        | Number    | -    | Primary Key           |
| <b>SalaryAmount</b> | Number    | -    | -                     |
| <b>PayDate</b>      | Date      | -    | Default SYSDATE       |
| <b>StaffID</b>      | Number    | -    | Foreign Key (StaffID) |

## 10. Administration

| Attribute         | Data Type | Size | Constraints                 |
|-------------------|-----------|------|-----------------------------|
| <b>AdminID</b>    | Number    | -    | Primary Key                 |
| <b>Name</b>       | Varchar2  | 30   | Not Null                    |
| <b>Contact</b>    | Varchar2  | 15   | Not Null                    |
| <b>Role</b>       | Varchar2  | 20   | Not Null                    |
| <b>ShowroomID</b> | Number    | -    | Foreign Key<br>(ShowroomID) |

## 11. Insurance

| Attribute                 | Data Type | Size | Constraints         |
|---------------------------|-----------|------|---------------------|
| <b>InsuranceID</b>        | Number    | -    | Primary Key         |
| <b>InsuranceProvider</b>  | Varchar2  | 30   | Not Null            |
| <b>InsuranceStartDate</b> | Date      | -    | -                   |
| <b>InsuranceEndDate</b>   | Date      | -    | -                   |
| <b>CoverageAmount</b>     | Number    | -    | -                   |
| <b>CarID</b>              | Number    | -    | Foreign Key (CarID) |

|            |        |   |                          |
|------------|--------|---|--------------------------|
| CustomerID | Number | - | Foreign Key (CustomerID) |
|------------|--------|---|--------------------------|

## **12.Booking**

| Attribute     | Data Type | Size | Constraints                                                   |
|---------------|-----------|------|---------------------------------------------------------------|
| BookingID     | Number    | -    | Primary Key                                                   |
| BookingDate   | Date      | -    | Default SYSDATE                                               |
| BookingStatus | Varchar2  | 20   | Check(BookingStatus IN ('Pending', 'Confirmed', 'Cancelled')) |
| AdvanceAmount | Number    | -    | -                                                             |
| CustomerID    | Number    | -    | Foreign Key (CustomerID)                                      |
| CarID         | Number    | -    | Foreign Key (CarID)                                           |

## **13.Warranty**

| Attribute         | Data Type | Size | Constraints         |
|-------------------|-----------|------|---------------------|
| WarrantyID        | Number    | -    | Primary Key         |
| WarrantyStartDate | Date      | -    | -                   |
| WarrantyEndDate   | Date      | -    | -                   |
| CoverageDetails   | Varchar2  | 100  | -                   |
| CarID             | Number    | -    | Foreign Key (CarID) |

# **(SQL Statements)**

- 1. Create Statement**
- 2. Insert Statement**
- 3. Delete Statement**
- 4. Update Statement**
- 5. Select Statement**

# **(Views Creation)**



*(Tables)*

